

Logarithm

Here $b > 0$, $b \neq 1$, and $x > 0$.

$$\log_b x = y \iff b^y = x$$

What quantity does the logarithm return?

Inverse functions

Assume $b > 0$ and $b \neq 1$.

$$f(x) = b^x \quad g(x) = \log_b x$$

What input restriction must g have?

From exponential to logarithmic

$$4^3 = 64$$

What is the equivalent logarithmic statement?

From logarithmic to exponential

$$\log_5 125 = 3$$

What is the equivalent exponential statement?

A negative logarithm

$$\log_2 \left(\frac{1}{8} \right)$$

What is the exact value?

A fractional logarithm

$$\log_9 3$$

What is the exact value?

A base below one

$$\log_{1/2} 8$$

What exponent gives the argument?

Logarithm of one

Assume $b > 0$ and $b \neq 1$.

$$\log_b 1$$

What is the value for every allowed base b ?

Logarithm of the base

Assume $b > 0$ and $b \neq 1$.

$$\log_b b$$

What is the value for every allowed base b ?

Undoing an exponential

$$\log_3(3^u)$$

What is the value for every real u ?

Undoing a logarithm

$$5^{\log_5 x}$$

What is the value, and what condition must x satisfy?

An undefined real logarithm

$$\log_4(-16)$$

Why is this expression undefined over the real numbers?

Zero as an argument

$$\log_7 0$$

Could any real exponent of 7 produce the argument?

Finding an argument

$$\log_3 x = -2$$

What positive value of x satisfies the equation?

Finding a base

$$\log_b 81 = 4$$

What allowed base b satisfies the equation?

Common logarithm

A common logarithm has base 10.

$$\log x = \log_{10} x$$

What is the exact value of $\log(0.001)$?

Approximating a common logarithm

$\log 35$

Between which consecutive integers does the value lie?

Natural logarithm

A natural logarithm has base e .

$$\ln x = \log_e x$$

What is the exact value of the natural logarithm of e cubed?

Calculator evaluation

In 7

What is the value rounded to four decimal places?

Exact and approximate values

$\ln 7$ 1.9459

Which expression is exact, and which is a rounded approximation?